Implementation

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Install vtamR and Third Party Programs - v0.3.2

Implementation

The heavy lifting of sequence analyses are done by the following third party programs. They should be installed on your computer:

- BLAST(Altschul et al., 1990 (https://pubmed.ncbi.nlm.nih.gov/2231712/)) (v2.11.0+)
- VSEARCH (Rognes et al., 2016 (https://peerj.com/articles/2584/)) (v2.7.0)
- CUTADAPT(Martin, 2011 (https://journal.embnet.org/index.php/embnetjournal/article/view/200/479)) (v4.0)
- SWARM (Mahé et al., 2015 (https://peerj.com/articles/1420/)) (v2.1.12)
- VSEARCH (optional)

`vtamR` has been tested using the above mentioned versions, but it should work with later versions.

`vtamR` was tested on **Windows** and **Linux**, and should work in all operating systems.

Install vtamR

Install `vtamR` and its CRAN dependencies by `pak` .

```
if(!requireNamespace("pak", quietly = TRUE)) install.packages("pak")
pak::pkg_install("meglecz/vtamR", dependencies=TRUE)
```

Alternatively, you can use `devtools` .

```
if (!requireNamespace("devtools", quietly = TRUE)) install.packages("devtools")
devtools::install_github("meglecz/vtamR", dependencies = TRUE)
```

Install Bioconductor packages manually if you intend to use `TaxAssignRDP` function.

```
# Install Bioconductor suggested packages manually if needed (only for TaxAssignRDP)
if (!requireNamespace("BiocManager", quietly = TRUE)) install.packages("BiocManager")
if (!requireNamespace("rRDP", quietly = TRUE)) BiocManager::install("rRDP")
if (!requireNamespace("rRDPData", quietly = TRUE)) BiocManager::install("rRDPData")
if (!requireNamespace("Biostrings", quietly = TRUE)) BiocManager::install("Biostrings")
```

Read the **Manual (https://people.imbe.fr/~emeglecz/vtamR)** on how to construct a full metabarcoding pipeline.

Install third party programs

Linux

- `VSEARCH` https://github.com/torognes/vsearch (https://github.com/torognes/vsearch)
- `BLAST` https://www.ncbi.nlm.nih.gov/books/NBK52640/ (https://www.ncbi.nlm.nih.gov/books/NBK52640/)
- `CUTADAPT` https://cutadapt.readthedocs.io/en/stable/installation.html (https://cutadapt.readthedocs.io/en/stable/installation.html)
- `SWARM` https://github.com/torognes/swarm (https://github.com/torognes/swarm)
- `VSEARCH` (optional) https://github.com/madler/pigz (https://github.com/madler/pigz)

Add the executables to your PATH

This step is optional. If third-party executables are included in your `PATH`, you won't need to manually specify their locations when running functions that use them.

If you prefer **not** to modify your `PATH`, be sure to provide the full path to each program using the corresponding `vtamR` function argument (e.g., `vsearch_path`).

1. Edit (or create) your `~/.Renviron` file

```
nano ~/.Renviron
```

2. Add the directory to the PATH

```
PATH="/home/user/tools/myprogram/bin:${PATH}"
```

Note: The PATH must contain a **directory**, not an executable file. For example, use: `/home/username/miniconda3/envs/vtamr/bin` and **not**: `/home/username/miniconda3/envs/vtamr/bin/cutadapt`

3. Restart R or RStudio

The new PATH is loaded at startup.

4. Verify that R sees the correct executables

```
system("cutadapt --version")
system("swarm --version")
system("vsearch --version")
system("blastn -version")
system("pigz --version")
```

Windows

Download binaries and save them to a convenient place on your computer (path without space, e.g. `C:/Users/Public/`)

VSEARCH

- Download binaries from https://github.com/torognes/vsearch/releases/tag/v2.23.0 (https://github.com/torognes/vsearch/releases/tag/v2.23.0)
- Decompress the zip file
- The executable are found in `vsearch-x.xx.x-win-x86_64/bin/`

CUTADAPT

- Download `cutadapt.exe` from https://github.com/marcelm/cutadapt/releases (https://github.com/marcelm/cutadapt/releases)

SWARM

- Download binaries from https://github.com/torognes/swarm/releases (https://github.com/torognes/swarm/releases)
- Decompress the zip file
- The executable is in the `swarm-x.x.x-win-x86_64/bin` directory

BLAST

- Download the installer (ncbi-blast-x.xx.x+-win64.exe) from: https://ftp.ncbi.nlm.nih.gov/blast/executables/LATEST/ (https://ftp.ncbi.nlm.nih.gov/blast/executables/LATEST/)
- Double-click the .exe file, accept the license agreement, and keep the suggested installation path. This will automatically add BLAST to your system PATH, so you won't need to specify its location when using it.
- The executables will be installed in: `C:\Program Files\NCBI\blast-x.xx.x+\bin`

VSEARCH (Optional)

- Download pigz-win32.zip (https://sourceforge.net/projects/pigz-for-windows/files/)
- Decompress the zip file
- The executable (pigz.exe) is in the pigz-win32 directory

Add executables to your PATH

This step is optional. If third-party executables are included in your system PATH, you won't need to manually specify their locations when running functions that use them.

If you prefer **not** to add them, make sure to provide the full path to each program through the appropriate `vtamR` function argument (e.g., `vsearch_path`).

You can use either **GUI** or **PowerShell** to add executables to your PATH.

1. Using the Windows GUI (System Settings)

- Press **Win + R**, type `sysdm.cpl` , and press **Enter**. *(Opens System Properties.)*
- Go to the **Advanced** tab.
- Click **Environment Variables…**
- In the **System variables** section, find and select **Path**.
- Click **Edit…**
- Click **New**, then paste the directory you want to add (for example: `C:\Users\YourName\Tools\bin`)
- Click **OK** on all windows to save.

2. Using PowerShell

- Press **Win + R**, type `powershell` , and press **Enter**. *(Opens Powershell)*

```
setx PATH "$($env:Path);C:\My\New\Folder\vsearch:C:\My\New\Folder\cutadapt;C:\My\New\Folde
```

- Close de Powershell, since the change applies only to **new** terminals, not the current one.

Verify Program Versions

Open a Command Prompt on Windows (**Win + R**, type `cmd`)

```
cutadapt --version
swarm --version
vsearch --version
pigz --version
blastn -version
```

Note *These commands will display the version of each program **if the executables are in your PATH**. Otherwise, don't forget to provide the full path to the executables.*

Example:

```
C:/Users/Public/vsearch-x.xx.x-win-x86_64/bin/vsearch --version
```

TaxAssign reference data base

TaxAssignLTG

For `TaxAssignLTG` (BLAST based Lowest Common Ancestor method), a ready-to-use COI database is available from OSF OSF (https://osf.io/vrfwz/), (Meglécz, 2023 (https://onlinelibrary.wiley.com/doi/10.1111/1755-0998.13756)).

Download and extract using:

```
library(vtamR)

download_osf(filename = "COInr_for_vtam_2025_05_23_dbV5.tar.gz",
    url = "https://osf.io/download/jyhz6/",
    dest_dir = "COInr_db",
    untar = TRUE,
    quiet = FALSE
    )
```

To check for newer versions:

- Go to: https://osf.io/vrfwz/files/osfstorage (https://osf.io/vrfwz/files/osfstorage)
- In **Files** open the the **dbV5** folder.
- Choose your version → Click on the tree-dot menu → Right-click Download → Copy Link Address.

Alternatively, you can also create a custom version from the **TSV format** database available at Zenodo (https://zenodo.org/records/15515860) and make a custom version using mkCOInr (https://github.com/meglecz/mkCOInr) (Meglécz, 2023 (https://onlinelibrary.wiley.com/doi/10.1111/1755-0998.13756)).

For other markers you will need a database formatted to BLAST (tutorial-vtamr-pipeline.html#reference-database-for-taxonomic-assignments), containing taxIDs and a taxonomy file (tutorial-vtamr-pipeline.html#reference-database-for-taxonomic-assignments). If all sequences are extracted from NCBI-nt, the taxonomy file provided in OSF (https://osf.io/vrfwz/) can be used without modification.

TaxAssignRDP

`TaxAssignRDP` function uses de RDP algorithm to assign bacterial 16S sequences. The database is provided in the `rRDPData` R data package.

It is also possible to provide to provide a trained classifier object created with `trainRDP()` .